

BRUTE FORCE ATTACK

INTRODUCTION:

In the field of cybersecurity, one of the most common and oldest attack techniques is the brute force attack. It involves systematically guessing credentials or encryption keys until the correct one is found. Despite its simplicity, it continues to be effective—especially against weak or reused passwords. This report explores the fundamentals of brute force attacks, the concept and role of wordlists in such attacks, methods to create customized wordlists, and the real-world impact these attacks have had in terms of data breaches and system compromises. Understanding these concepts is essential for ethical hackers and security analysts working to secure systems against such threats.

1. What is a Brute Force Attack?

A Brute Force Attack is a hacking technique where the attacker tries all possible combinations of credentials or keys until the correct one is found.

□ Types of Brute Force Attacks:

- Simple Brute Force: Attempts every possible password combination.
- Dictionary Attack: Uses pre-made lists of common passwords.
- Hybrid Attack: Combines dictionary words with random characters.
- Credential Stuffing: Uses leaked credentials across multiple platforms.
- Reverse Brute Force: Starts with a known password and tries it against many usernames.

□ Example:

Trying admin: **admin**, admin: **123456**, admin: **password**, etc. until login is successful.

□ Used Against:

- SSH/FTP/SMB logins

- Web forms (WordPress, phpMyAdmin, etc.)
- Wi-Fi passwords (WPA/WPA2)

2. What is a Wordlist?

A wordlist is a file containing potential passwords or phrases used during brute force or dictionary attacks.

☐ Sources of Wordlists:

- Publicly leaked data (e.g., LinkedIn, Adobe)
- Human behavior analysis (common patterns)
- Custom generation (based on target info)

☐ Famous Wordlists:

- rockyou.txt (contains 14+ million leaked passwords)
- SecLists (tool-specific lists from GitHub)
- Weakpass (huge collection of breached credentials)

3. How to Prepare a Wordlist?

- ☐ Use tools like Crunch to generate custom password lists based on patterns.
- ☐ Use Cewl to scrape keywords from target websites and create a smart wordlist.
- ☐ Manually create lists using personal information (name, DOB, pet name, etc.).
- ☐ Combine known passwords with rules using John the Ripper (--rules flag).
- ☐ Use leaked wordlists like rockyou.txt as a base and filter relevant entries.
- ☐ Merge multiple lists and remove duplicates using:

```
cat list1.txt list2.txt | sort -u > final.txt
```

- ☐ Use social engineering to guess likely words and phrases used by the target.
- ☐ Tailor wordlists based on location, language, job role, or interests.

4. Total Breaches Caused by This Attack

Brute-force attacks have caused several high-profile breaches, especially when users reuse weak passwords or services lack multi-factor authentication.

Recent Stats (as per industry reports):

- Over 80% of hacking-related breaches in 2023 involved brute force or stolen credentials (Verizon DBIR).
- Credential stuffing (a brute-force variant) led to billions of login attempts per month in large corporations.
- Attacks like BlueKeep (RDP brute force) and Microsoft Exchange brute force exposed thousands of enterprise servers.

Companies now mitigate these with:

- Rate limiting
- Multi-Factor Authentication (MFA)
- Login attempt monitoring
- CAPTCHAs and account lockouts

Here are some companies/organisations which are affected by this brute force attack:

1. Facebook (Meta)

- Suffered credential stuffing attacks using brute-forced or leaked passwords.

2. GitHub

- Attackers used brute-force attempts and credential stuffing with compromised accounts.

3. Zoom

- Faced large-scale credential stuffing attacks early in the pandemic via brute-forced credentials.

4. Microsoft Exchange Servers

- Targeted in brute-force campaigns against exposed RDP and email login portals.

5. Adobe Systems

- In a major breach, brute force techniques were used on weakly hashed passwords.

CONCLUSION:

Preparing a powerful wordlist is more than just collecting random passwords — it's about understanding the target, using the right tools, and thinking creatively. Tools like Crunch, Cewl, and John the Ripper help automate this process, but the real effectiveness comes from customizing the list based on social and contextual intelligence. With a well-prepared wordlist, ethical hackers can strengthen their testing capabilities and identify weak credentials before malicious actors do.